

Agrilus planipennis (Emerald Ash Borer)

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The emerald ash borer (*Agrilus planipennis Fairmaire*) is a half-inch long metallic green beetle originally from Asia that can be found in nearly every county of the Commonwealth.

It was first identified in North America during 2002 and in western Pennsylvania during 2007.

The larval stage of this beetle is harmful, feeding exclusively on ash trees under the bark and killing them three to five years after infestation.

Signs and symptoms of an emerald ash borer infestation include:

- Upper crown dieback
- Epicormic branching
- Bark splits
- Bark flaking
- Tissue damage resulting from woodpecker predation
- D-shaped adult beetle exit holes in the bark
- S-shaped larval feeding galleries just below the bark

All native North American ash species, ash cultivars, and the white fringetree are susceptible to emerald ash borer.

Management of Emerald Ash Borer in Pennsylvania Forests

Emerald ash borer is a serious threat to the 308 million ash trees in the forests of Pennsylvania, including:

- Pumpkin ash -- a state species of concern
- Ash seed orchards managed by DCNR's Bureau of Forestry
- White ash, green ash, black ash, blue ash, and the white fringetree, a species in the same taxonomic family as ash.

Without active management, it is predicted that emerald ash borer will severely decimate populations of ash trees in the state.

As of 2014, ash forests in Pennsylvania have been reduced by 12 percent.

To address the immediate and long-term impact of emerald ash borer on state forest lands in Pennsylvania, the bureau has developed a comprehensive ash management plan, [Ash Management in State Forest Lands Under Pressure from the Emerald Ash Borer \(PDF\)](#), with the following objectives:

- Managing ash as a component in the forest
- Protecting endangered ash species
- Mitigating potential negative impacts
- Conserving economic value through silviculture
- Managing seed orchards and collecting seeds
- Conducting training and public outreach

Currently, there are three control options for the emerald ash borer:

- Tree removal
- Chemical control
- Biological control

Tree Removal

Harvesting ash trees reduces the amount of phloem (innermost layer of the bark) available to developing emerald ash borer larvae, which in term slows their population growth and spread.

Chemical Control

Field surveys are conducted within the state forests to determine the number of ash trees that have a significant value within a forest district.

Ash trees that have a historic or ecological significance as well as select seed producing ash trees are currently being treated with a systemic insecticide to prevent attack by emerald ash borer.

In predetermined locations, the bureau has treated 1,700 ash trees within the state forest system for emerald ash borer between the years 2014–2015.

Biological Control and Host Tolerance

The long-term management of emerald ash borer will depend on the success of classical biological control -- the intentional introduction and permanent establishment of exotic agents for long-term pest control, and through a tree breeding program to select for ash tolerance to emerald ash borer attack.

The bureau is actively working with USDA Forest Service researchers to identify ash trees that are tolerant or resistant to emerald ash borer attacks.

DCNR's Bureau of Forestry is working on establishing a biological control program for the pest. The following four parasitoid species are currently being utilized for emerald ash borer biological control in North America:

- Oobius agrili
- Tetrastichus plannipenssi
- Spathius agrili
- Spathius galinae

Tips for Homeowners With Emerald Ash Borer Infestations

Several insecticide options are available to effectively treat landscape ash trees threatened by the emerald ash borer; however, controlling insects that feed under the bark with insecticides can be challenging.

To prevent an emerald ash borer infestation requires some care when selecting an insecticide product and application method to ensure the product is applied at the proper rate and time.

Insecticides that can effectively control emerald ash borer include:

- Systemic insecticides that are applied as soil injections or drenches
- Systemic insecticides applied as trunk injections
- Systemic insecticides applied as lower trunk sprays
- Protective cover sprays that are applied to the trunk, main branches, and foliage

Some are marketed for use by homeowners, while the most effective option is intended for use only by certified pesticide applicators due to the use of a Restricted Use Pesticide.

Insecticide applications should begin before your ash tree is attacked by emerald ash borer.

Only ash trees that are healthy and are of value to the homeowner should be considered for treatment.

Other ash trees should be removed before they become a hazard.

Additional Resources

[An Introduction to the Emerald Ash Borer Community Management Plan \(PDF\)](#)

[A Management Plan for Pennsylvania Communities \(PDF\)](#)

[Emerald Ash Borer Management Plan Template \(PDF\)](#)

[The Community Forest Inventory Decision Model \(PDF\)](#)

[A Software Tool For Assessment -- iTree \(PDF\)](#)

[Landscape Tree Appraisal \(PDF\)](#)

[Insecticide Options \(PDF\)](#)

[Role of BioControl in Protecting Ash Regeneration \(PDF\)](#)

[Potential Side Effects of Emerald Ash Borer Insecticides \(PDF\)](#)

[Urban Wood Utilization Planning \(PDF\)](#)

[Emerald Ash Borer Treatment Costs -- DCNR Recommendations \(XLS\)](#)

[Emerald Ash Borer Infestation Map \(PDF\)](#)

EAB Management Plans Completed

[City of Philadelphia \(PDF\)](#)

[City of Easton \(PDF\)](#)

[Greenwood Furnace State Park \(PDF\)](#)

[City of Lancaster \(PDF\)](#)

[Lewisburg Borough \(PDF\)](#)

[City of Pottsville \(PDF\)](#)

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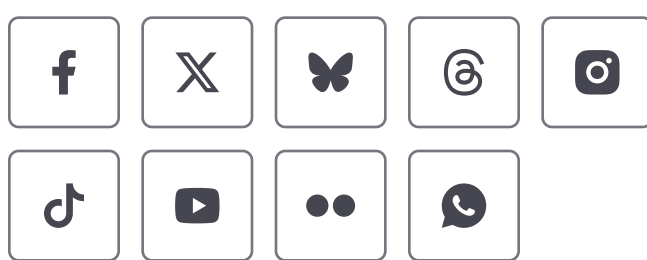
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